

CATEGORY		LOW-END INTEL MOTHERBOARDS					
	Brand	Jetway	Mercury	Gigabyte	Gigabyte	Mercury	Gigabyte
	Model	P4MDUT	PVM266AM-L	GA-VM533	GA-8S650GXM-P	PS650GXM-L	GA-8S648FX
	Chipset	VIA P4M266A	Via P4M266A	VIAP4M533	SIS 650GX	SIS 650GX	SiS 648FX
	Interface type	Socket 478	Socket 478	Socket478	Socket 478	Socket 478	Socket 478
FEATURES (20%)	CPU Speed (GHz) / FSB supported (MHz)	P4 3.06 / 533	P4 3.06 / 533	P4 3.06 / 533	P4 3.06 / 533	P4 3.06 / 533	P4 3.2 / 800
	Memory Speed (MHz) / Memory slots (nos)	DDR 266 / 2	DDR 266 / 2	DDR 266 / 2	DDR 333 / 2	DDR 266 / 2	DDR 400 / 2
	Form Factor	Micro ATX	Micro ATX	Micro ATX	Micro ATX	Micro ATX	Micro ATX
	AGP slot / Onboard AGP	✓/✓	✓/✓	✓/✓	✓/✓	✓/✓	✓/✗
	PCI Slots	3	2	3	3	3	5
	Onboard Sound	✓	✓	✓	✓	✓	✓
	USB ports / Rear/ Ports Via Cable	6/2/4	6/4/2	6/2/4	6/2/4	6/2/4	6/2/4
	Ethernet	✗	✓	✓	✓	✓	✓
	Hyperthreading Support	✓	✗	✓	✓	✗	✓
	Core voltage adjustment	✗	✗	✓	✓	✗	✓
	Frequency adjustment	✗	✗	✓	✓	✗	✓
	Component layout on the motherboard (S05)*	1.5	2.75	3	2.75	2	2.75
	Package contents						
	Manual / CDs	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓
	Necessary cables provided	✓	✓	✓	✓	✓	✓
PERFORMANCE (80%)	Gaming Benchmarks: Serious Sam Second Encounter						
	Average fps	25.95	21.73	25.67	17.93	15.10	18.10
	Quake III Arena						
	Average fps	30.37	23.20	32.23	43.33	36.00	42.10
	Real World Benchmarks						
	ZDbench Content Creation 2004	20.1	19.5	20.4	22	22.1	25.5
	ZDbench Business Winstone 2004	14.5	14.5	14.9	18.4	18.8	21.4
	Video Encoding (Time in secs)	112	124	111	106	153	89
	Synthetic Benchmarks						
	3D Mark 2001 SE (patched for onboard without AGP slot)	660	500	695	1,138	1,003	1,123
	PCMark 04						
	CPU Index	3,724	3,839	3,375	3,594	3,425	4,112
	Memory Index	1,667	1,585	1,590	2,314	2,248	2,359
	Graphics	245	194	252	528	514	512
	Hard disk index	2,931	2,970	2,995	3,267	3,345	3,342
	Total Score	2,630	2,451	2,604	2,759	2,696	3,822
	Sisoft Sandra 2004 Professional: CPU sub-system						
	CPU Dhrystone/Whetstone	7,856/ 2,256	7,696/ 2,245	7,815/ 2,233	7,139/ 2,069	7,176/2,056	7,892/2,271
	Multimedia - CPU Integer / FPU SSE	18,958/22,411	18,835/22,313	18,767/22,095	17,369/20,506	17,293/20,491	19,105/22,506
	Memory sub-system						
	ALU / RAM	1,097	951	1,191	1,631	1,559	2,490
	FPU / RAM	1,095	944	1,186	1,633	1,563	2,495
	Disk drive						
	Drive Index MBps	20,538	20,778	20,628	30,680	26,196	26,389
OVERALL SCORE	Performance	36.42	34.94	37.68	43.44	40.42	48.66
	Features	8.90	9.65	12.80	13.65	9.20	16.65
	Grand Total (Performance + Features)	45.32	44.59	50.48	57.09	49.62	65.31
	VFM (Index)	17.10	18.97	17.41	19.35	18.38	14.51
	Price (Rs)	2,650	2,350	2,900	2,950	2,700	4,500
Grade		B+	B+	B+	B+	B+	B

*Scale of 5



The features offered with the Mercury PS650GXM-L suited its price very well

ic cores: Intel's onboard graphical controller is much better than the S3 core used by VIA and SiS—the image was vibrant, and the texture, true with the Intel Extreme graphics; all these aspects were completely washed out on the S3-based boards.

Conclusion

Gigabyte's GA-8I865GVMK completely dominated the performance arena raking in great scores in all the tests. The only gripe with this board is the lack of an AGP slot. Priced at Rs 5,500, (Rs 600 more than the next-best in terms of performance), this board took the Best Performance title with ease. If you find the price a little too high, your next choice should be the Gigabyte GA-8GEM667, or HIS M53-1A, for their good performance and affordable pricing.

When it comes to the Value for Money award, nothing could beat the Mercury Pi845GLM-L, which performs acceptably, at an amazing price of just Rs 2,625.